

Rahul Babu

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SUMMARY

Software Engineer experienced in full-stack and backend systems and data-driven applications, with strengths in Agentic AI, LLM orchestration, and ML/NLP pipelines for research and analytics workflows. Built production AI solutions in Python, JavaScript, and React with cloud deployments on AWS and GCP with CI/CD.

EDUCATION

Arizona State University, Tempe, AZ - M.S. in Computer Science

Aug 2023 - May 2025

Graduate research in AI and Data Science. Coursework: Statistical Machine Learning, Data Mining, Distributed Database Systems.

AWS Certified Solutions Architect - Associate [Verify](#)

June 2025

EXPERIENCE

Friedman Vartolo LLP | Software Engineer | New York, NY (Remote)

May 2026 - Current

- Engineered a suite of internal full-stack apps (React, TypeScript, Bun, Python) for the legal team; case management, content creation, legal research - deployed on Azure with Doppler and GitHub CI/CD.
- Built DCO (Direct Client Operations), an app powered by the Anthropic API that turns case details into a tailored PDF brief, slide deck, cheat sheet, and client email customized to the case's state, with built-in version control and inline editing.
- Built a Calendar application in React and TypeScript to manage attorney assignments, case appearances, docket numbers, and client data, with automated email generation and dynamic data formatting for legal teams.
- Architected a Law Tracker application that aggregates real-time news and data from multiple external APIs to monitor policy changes, new legislation, and legal developments relevant to the firm.
- Developed a Graphic Design Automation app enabling the marketing team to create and schedule LinkedIn posts programmatically, reducing manual content-production time through templated, automated output.

Street Care | Full Stack Developer | San Francisco, CA (Remote)

Dec 2025 - July 2026

- Optimized Firestore read and write operations through batching and caching strategies across React workflows, cutting page load times from 3.8s to 1.5s and reducing database costs by 15% across 200+ monthly user requests.
- Refactored monolithic React components into a modular architecture with code splitting, adding new pages and feature modules, reducing JavaScript bundle size by 35% and improving maintainability across 15+ modules. *(Live)*

Hovian Inc. | Software Engineer Intern | Cucamonga, CA (Remote)

July 2025 - Dec 2025

- Developed a React + Electron desktop application, integrating REST APIs and WebSockets to enable real-time Outlook and system notifications, along with AI-generated response workflows.
- Built and maintained the desktop UI (JavaScript, HTML5, CSS3), shipping 10+ reusable renderer views (chat, settings, toggles) with tabbed/multi-view navigation for notification and workflow management.

ASU Decision Theater Network | AI Project Aide | Tempe, AZ (Hybrid)

May 2024 - May 2025

- Performed geospatial opioid risk analysis using PostGIS and spatial SQL, and developed an ML pipeline in Python and R using XGBoost and ensemble methods on 50K+ census records to generate regional risk predictions.
- Delivered a real-time risk dashboard (React + TypeScript) backed by FastAPI services, containerized with Docker and deployed on Google Cloud Run for low-latency filtering. *(Live)*
- Deployed a RAG-based LLM chatbot using Llama 3, LangChain, and Chroma vector search on Jetstream2 HPC to summarize 100+ interview transcripts for Maricopa County and support ASU's LLM resilience research. *(Live)*

INNOHAT Systems | Software Developer | Chennai, TN, India

Jan 2022 - July 2023

- Designed hybrid storage using PostgreSQL for transactions and MongoDB for product metadata and reviews, reducing schema-change overhead while keeping financial data ACID-compliant.
- Orchestrated Python ETL pipelines (pandas, SQLAlchemy, PyMongo) merging order history with review-sentiment data, cutting pipeline runtime by 40% across 20M+ rows and 5M+ documents.
- Containerized services with Docker and deployed on AWS (EC2, RDS, S3, DocumentDB), with CloudWatch monitoring for stable performance under seasonal traffic spikes.

PROJECTS

ForecastHub | AI Weather Intelligence Platform - React, FastAPI, PostgreSQL, MLflow, Vercel, Render

Live: [forecast-hub-ai.vercel.app](#) | Github: [github.com/rahul-1415/forecast-hub](#)

- Built a multi-dashboard weather intelligence platform (React, FastAPI, PostgreSQL) with OpenAI-generated clothing, timing, and health suggestions, and automated MLOps via GitHub Actions + MLflow for hourly ingestion, daily model training, and deployment across Vercel and Render.

Loadouts | Social Media Platform for Tech Collections - React, Next.js, TypeScript, Tailwind CSS, Supabase, Vercel

Live: [loadouts-demo.vercel.app](#) | Github: [github.com/rahul-1415/loadouts](#)

- Built a Next.js full-stack social platform for curated tech collections with SSR, dynamic routing, and Supabase auth (Google/GitHub OAuth), real-time data flows, and edge-deployed APIs for profiles, publishing, and category feeds.

SKILLS

Backend & APIs: Python (Pandas, NumPy, async), Django, Flask, FastAPI, Java, REST APIs, Microservices, SQL/PostgreSQL/MySQL.

Frontend & Full-Stack: React, Angular, TypeScript, JavaScript (ES6+), HTML5/CSS3, Node.js, Express, GraphQL, Redux, Webpack.

ML & AI: LLMs (GPT, LLaMA), LangChain, RAG, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers; AI-assisted dev: Cursor, Claude Code, GitHub Copilot, Kiro.

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB, ECS, CloudWatch, CloudFormation, SQS/SNS), Terraform, Docker, Kafka, CI/CD, Git/GitHub/GitLab.